141. The decrease in atomic volume from Cr to Cu is very negligible because

(a) Increase in nuclear change
(b) Screening effect
(c) Unpaired electrons of Cr
(d) None

Answer: (b) Screening effect

Explanation: The correct choice is Screening effect. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

142. The heaviest atom amongst the following is

(a) Uranium
(b) Radium
(c) Lead
(d) Mercury

Answer: (a) Uranium

Explanation: Compare the relevant atomic masses and/or densities; the listed correct option has the greatest value among the choices.

143. Thallium shows different oxidation states because

(a) It is a transition metal
(b) Of inert-pair effect
(c) Of its high reactivity
(d) Of its amphoteric character

Answer: (b) Of inert-pair effect

Explanation: Assign oxidation numbers from charge balance and the accessible ns/(n-1)d valence electrons of the transition element.

144. The test of ozone O3 can be done by

(a) Ag
(b) Hg
(c) Au
(d) Cu

Answer: (b) Hg

Explanation: The correct choice is Hg. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

145. Which of the following set of elements does not belong to transitional elements set

(a) Fe, Co, Ni
(b) Cu, Ag, Au
(c) Ti, Zr, Hf
(d) Ga, In, Tl

Answer: (d) Ga, In, Tl

Explanation: The correct choice is Ga, In, Tl. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

146. The transition metals mostly are

(a) Diamagnetic

(b) Paramagnetic
(c) Neither diamagnetic nor paramagnetic
(d) Both diamagnetic and paramagnetic

Answer: (b) Paramagnetic

Explanation: The correct choice is Paramagnetic. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

147. The correct statement in respect of d-block elements

(a) They are all metals
(b) They show variable valency
(c) They form coloured ions and complex salts
(d) All above statements are correct

Answer: (d) All above statements are correct

Explanation: The correct choice is All above statements are correct. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

148. Which one of the following is an example of non-typical transition elements

(a) Li, K, Na
(b) Be, Al, Pb
(c) Zn, Cd, Hg
(d) Ba, Ca, Sr

Answer: (c) Zn, Cd, Hg

Explanation: The correct choice is Zn, Cd, Hg. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

149. Which one is wrong in the following statements

(a) Gold is considered to be the king of metals
(b) Gold is soluble in mercury
(c) Copper is added to gold to make it hard
(d) None of these

Answer: (d) None of these

Explanation: The correct choice is None of these. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

150. The number of unpaired electrons in Cr3+ ion is

(a) 3
(b) 5
(c) 4
(d) 1

Answer: (a) 3

Explanation: Determine the d-electron count of the atom/ion and apply Hund's rule. More unpaired electrons give a larger spin-only magnetic moment; d0 and d10 species are diamagnetic.

151. The metal ion which does not form coloured compound is

(a) Chromium

(b) Manganese

(c) Zinc

(d) Iron

Answer: (c) Zinc

Explanation: Partly filled d orbitals allow electronic transitions that absorb visible light; d0 or d10 ions are generally colourless unless charge-transfer effects operate.

152. Super alloys are usually

(a) Iron based

(b) Nickel based

(c) Cobalt based

(d) Based on all of these

Answer: (d) Based on all of these

Explanation: The correct choice is Based on all of these. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

153. The transitional metal which shows oxidation state from +2 to +7 belong to group

(a) VII B

(b) VI B

(c) II B

(d) III B

Answer: (a) VII B

Explanation: Assign oxidation numbers from charge balance and the accessible ns/(n-1)d valence electrons of the transition element.

154. Which of the following may be colourless

(a) Cr^{+3}

(b) Cu^{+}

(c) Fe^{+3}

(d) Cu^{2+}

Answer: (b) Cu^{+}

Explanation: Partly filled d orbitals allow electronic transitions that absorb visible light; d0 or d10 ions are generally colourless unless charge-transfer effects operate.

155. Which of the following ions is paramagnetic

(a) Cu^{+}

(b) Zn^{+2}

(c) Ti^{+3}

(d) Ti^{+4}

Answer: (c) Ti^{+3}

Explanation: Determine the d-electron count of the atom/ion and apply Hund’s rule. More unpaired electrons give a larger spin-only magnetic moment; d0 and d10 species are diamagnetic.

156. Which of the following metals absorbs hydrogen

(a) K
(b) Al
(c) Zn
(d) Pd

Answer: (d) Pd

Explanation: The correct choice is Pd. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

157. Which of the following ions is coloured

(a) Cu+
(b) Cu2+
(c) Ti4+
(d) V5+

Answer: (b) Cu2+

Explanation: Partly filled d orbitals allow electronic transitions that absorb visible light; d0 or d10 ions are generally colourless unless charge-transfer effects operate.

158. The metal present in B12 is

(a) Magnesium
(b) Iron
(c) Cobalt
(d) Manganese

Answer: (c) Cobalt

Explanation: The correct choice is Cobalt. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

159. Which metal does not give the following reaction M + water or steam →oxide + H

(a) Mercury
(b) Iron
(c) Sodium
(d) Magnesium

Answer: (a) Mercury

Explanation: The correct choice is Mercury. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

160. Ionisation potential values of d-block elements as compared to ionization potential value of f-block elements are

(a) Higher
(b) Equal
(c) lower
(d) All of these

Answer: (a) Higher

Explanation: Compare the stability of the resulting electronic configurations as well as effective nuclear charge; half-filled and fully filled subshells often produce anomalies.